# MATH 101 Unit 3: Polynomial, Rational, and Radical Functions

Practice 3 | Version 202610122020.42 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

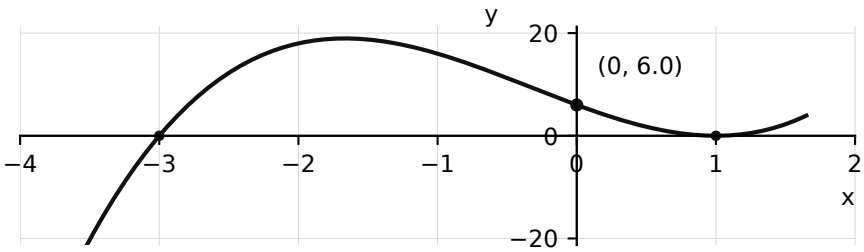
| QUESTIONS | ANSWERS |
|---|---|
| 1. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain.<br>$f(x) = 3\sqrt{x+9} - 12$ | 1. (a) Domain 4 pts<br>(a) Range 4 pts<br>(b) x-intercept 3 pts<br>(b) y-intercept 3 pts<br>(c) Infinite end 4 pts<br>(c) Finite end 2 pts |
| 2. Find (a) the vertex and the minimum or maximum value (state which); (b) all zeros.<br>$f(x) = 3(x-1)^2 - 48$ | 2. (a) Vertex 6 pts<br>(a) Minimum 4 pts<br>(b) Zeros 10 pts |
| 3. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts.<br>$f(x) = \frac{(x-3)(x+5)}{(x-7)(x-3)}$ | 3. (a) Domain 4 pts<br>(b) Hole 4 pts<br>(b) Vertical 3 pts<br>(b) Horizontal 3 pts<br>(c) x-intercept 3 pts<br>(c) y-intercept 3 pts |

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| 4. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions.<br>  | 4. Polynomial<br>16 pts<br>Left end<br>2 pts<br>Right end<br>2 pts |
| 5. Find all zeros: (a) list Rational Root Theorem candidates, then test only 1 and -1; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros.<br>$p(x) = x^3 - 4x + 3$ | 5. (a) Candidates/root<br>5 pts<br>(b) Quotient<br>7 pts<br>(c) Other zeros<br>8 pts |

# Instructor answer key: MATH 101 Unit 3: Polynomial, Rational, and Radical Functions

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

## TEST PAGE 1

1. (a) Domain
$[-9, \infty)$ — 4 pts

(a) Range
$[-12, \infty)$ — 4 pts

(b) x-intercept
$(7,\ 0)$ — 3 pts

(b) y-intercept
$(0,\ -3)$ — 3 pts

(c) Infinite end
$x \to \infty:\ f(x) \to \infty$ — 4 pts

(c) Finite end
$x \to -9^{+}:\ f(x) \to -12$ — 2 pts

2. (a) Vertex
$(1,\ -48)$ — 6 pts

(a) Minimum
$-48$ — 4 pts

(b) Zeros
$x = -3,\ 5$ — 10 pts

3. (a) Domain
$\mathbb{R} \setminus \{3, 7\}$ — 4 pts

(b) Hole
$(3,\ -2)$ — 4 pts

(b) Vertical
$x = 7$ — 3 pts

(b) Horizontal
$y = 1$ — 3 pts

(c) x-intercept
$(-5,\ 0)$ — 3 pts

(c) y-intercept
$(0,\ -\frac{5}{7})$ — 3 pts

## TEST PAGE 2

4. Polynomial
$2(x+3)(x-1)^2$ — 16 pts

Left end
$x \to -\infty:\ f(x) \to -\infty$ — 2 pts

Right end
$x \to \infty:\ f(x) \to \infty$ — 2 pts

5. (a) Candidates/root
$\{-3, -1, 1, 3\};\ x = 1$ — 5 pts

(b) Quotient
$x^2 + x - 3;\ R = 0$ — 7 pts

(c) Other zeros
$x = -\frac{1}{2} + \frac{\sqrt{13}}{2},\ \ -\frac{\sqrt{13}}{2} - \frac{1}{2}$ — 8 pts